



2.2. Closest Pair of Points Problem

"Divide, Conquer, and Connect the Closest"

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2. Divide and Conquer

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Every problem in computer science usually begins with a simple question. And honestly, this one sounds deceptively simple:

"Among all given points... which two are the closest together?"

At first glance, it almost feels trivial. Just calculate every distance and pick the smallest one, right? It is plausible. But the moment the number of points becomes large, something interesting happens. The problem suddenly becomes expensive. Very expensive.

Imagine trying to monitor thousands of airplanes in the sky, millions of GPS locations, or massive collections of objects in a simulation. Calculating every possible pair quickly becomes inefficient.

And this is where computational geometry starts becoming clever. Instead of comparing everything blindly, we begin asking a better question:

“Can we organize the problem in a smarter way?”

This is exactly the philosophy behind the **Closest Pair of Points Problem**. Rather than brute-forcing every possible distance, we use one of the most important strategies in algorithm design: **Divide and Conquer**.

The idea is elegant: Divide the points into smaller groups → Solve the problem inside each group → Then combine the results intelligently

And surprisingly, this reduces the complexity dramatically.

So in this section, we will not just solve the problem. We will explore how divide and conquer transforms what initially looks like a geometric problem, into a beautifully structured algorithmic strategy.

1. Problem Definition

Now let's properly define the problem. Suppose we are given n points on a 2D plane. Each point has an x -coordinate and a y -coordinate.

Our goal is simple: ***Find the pair of points with the smallest Euclidean distance between them.***

In other words, among all possible pairs of points, we want the two points that are physically closest together. This problem appears in many real-world applications. For example:

- In air traffic control, we may want to detect airplanes that come dangerously close to one another.
- In GPS and mapping systems, we may need to identify nearby locations efficiently.
- In computer graphics and simulations, collision detection often depends on finding nearby objects quickly.
- In clustering and data analysis, measuring proximity between data points becomes extremely important.

So while the problem sounds mathematical, it is also highly practical. To measure the distance between two points p and q , we use the Euclidean distance formula:

$$||p, q|| = \sqrt{(p_x - q_x)^2 + (p_y - q_y)^2}$$

This formula simply measures the straight-line distance between the two points. Now here's the interesting part. A straightforward solution would be:

- ***Compare every point with every other point → Compute all distances → Select the minimum distance***

And yes, this works. But there is a problem. If we have n points, then the number of comparisons becomes extremely large as n increases. The brute-force approach requires checking: $\frac{n(n-1)}{2}$ possible pairs.

That means the algorithm grows roughly proportional to: $O(n^2)$. Which becomes inefficient for large datasets. So instead of checking everything blindly, we will apply the divide and conquer strategy.

The main idea is: ***Divide the set of points into smaller subsets → Solve the closest-pair problem recursively inside each subset → Then combine the solutions carefully to find the global closest pair.***

And this clever strategy reduces the complexity significantly.

So the Closest Pair of Points Problem is not just about geometry. It is about learning how algorithmic structure can dramatically improve efficiency.

2. A Practical Example

Now let's finally see how the Divide and Conquer strategy actually works in practice.

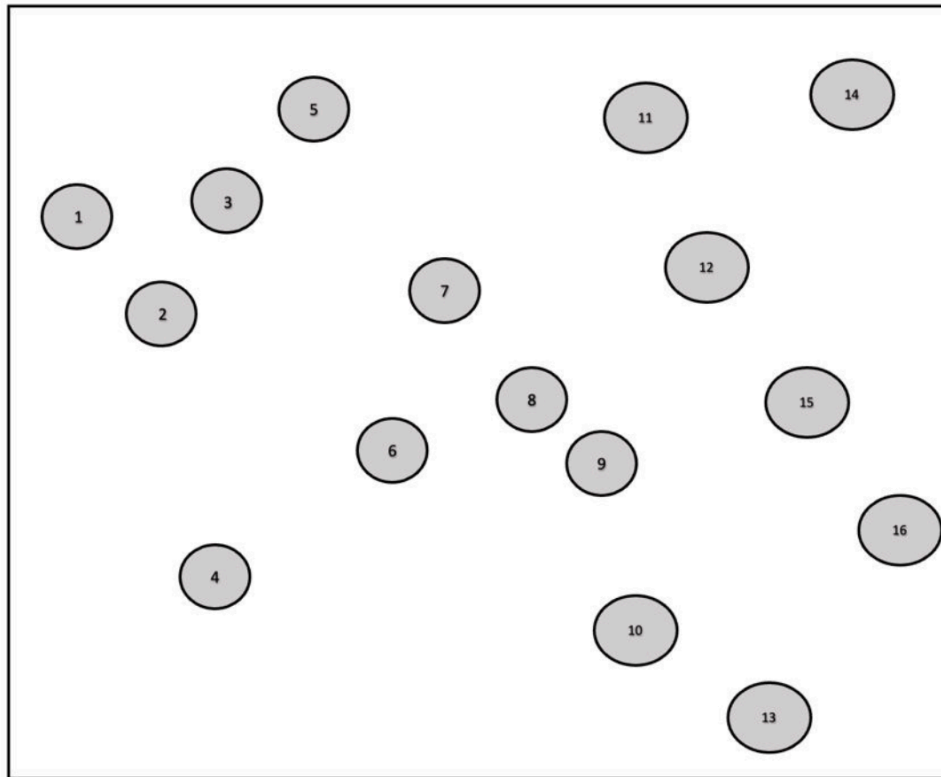
At first, the Closest Pair problem may seem like a purely geometric problem.

But underneath... it behaves very similarly to many recursive algorithms we've already studied.

The key idea is simple:

Instead of comparing every point with every other point, we divide the plane into smaller regions... solve them independently... then combine the results intelligently.

Consider the plane below.



We first convert all points into an array P of length n , sorted according to the x-coordinate.

So instead of thinking visually first, we think algorithmically:

$$P = [P_1, P_2, P_3, \dots, P_n]$$

Once the points are sorted, we locate the middle point:

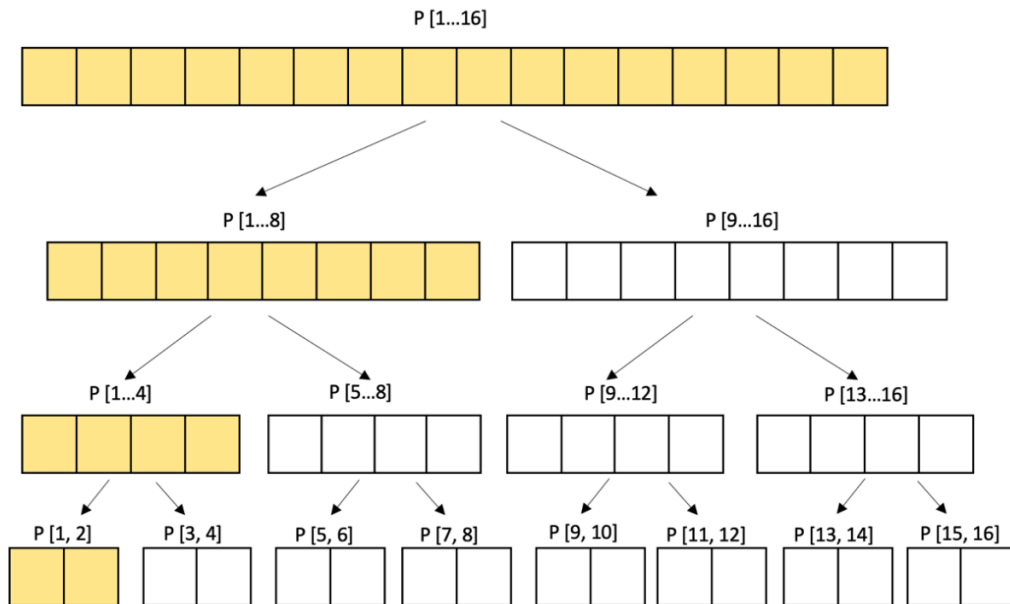
$$P[n/2]$$

This middle point acts like the dividing line of the plane.

Now we split the array into two smaller sub-arrays:

- Left sub-array: $P[0 \dots n/2]$
- Right sub-array: $P[n/2 + 1 \dots n - 1]$

Geometrically, this is equivalent to dividing the plane vertically into two halves. And now the Divide and Conquer process begins.

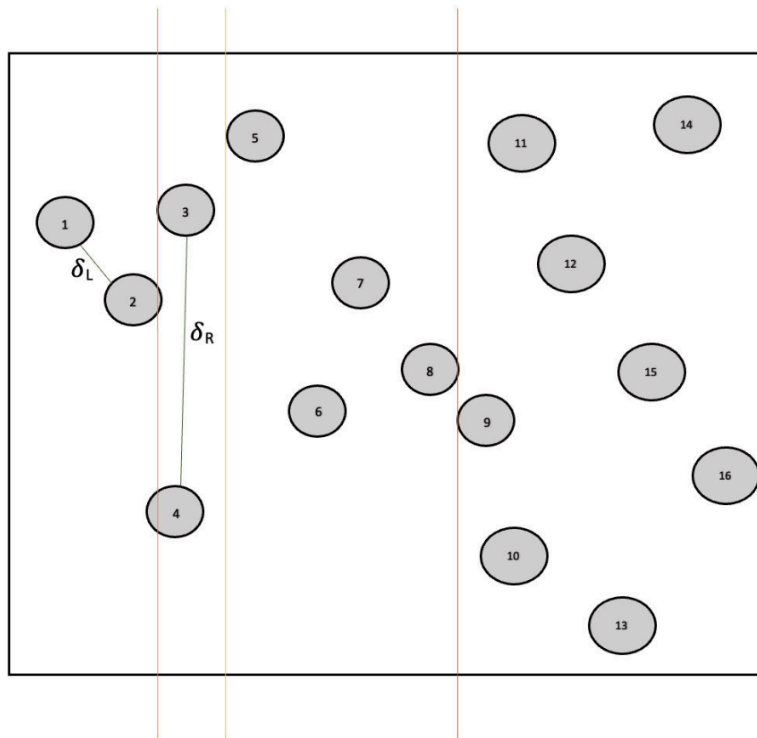


Using recursion, we independently search for the closest pair in both halves. This gives us:

- δ_L : the minimum distance in the left half
- δ_R : the minimum distance in the right half

Initially, suppose we examine a smaller example involving four points. To simplify the explanation, imagine that we temporarily ignore the other points and only focus on: P_1, P_2, P_3, P_4 . So after dividing them into two halves:

- Closest pair on the left side: $P[1, 2]$
- Closest pair on the right side: $P[3, 4]$



This image is used for demonstration only. Here, we only consider the four points: [P1, P2, P3, P4]

In the actual plane:

- The actual closest pair on the left side is P[6, 8]
- The actual closest pair on the right side is P[15, 16]

The result might vary depending on your perspective (some might argue that P[1, 2] is actual closest pair on the left side, which is valid).

We define: $\delta = \min(\delta_L, \delta_R)$. This means δ becomes the shortest distance we currently know. At first glance, this feels complete.

But there is a problem. A very important problem: **The Missing Case.**

And this is why I purposely only considered four points earlier—to clearly demonstrate that dividing the plane into two halves does not automatically guarantee that the closest pair has already been found.

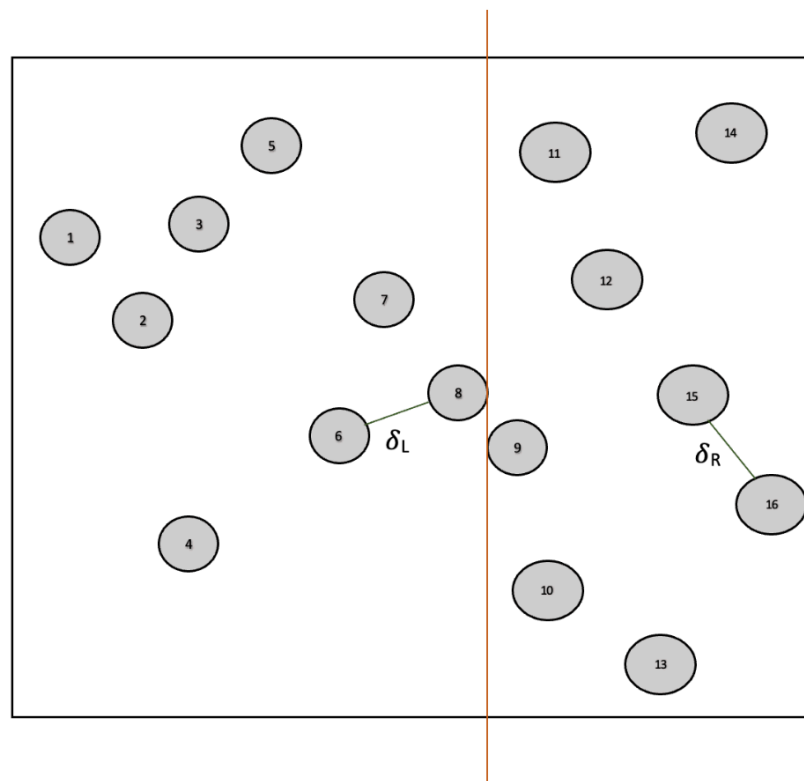
The recursive division only checks points inside their own halves. That means some possible pairs are completely ignored. For example: $P[1, 3]$ and $P[2, 4]$

These pairs lie across the dividing line. One point belongs to the left half. The other belongs to the right half. And one of these cross-border pairs could actually be the true closest pair.

So if we stop here, the algorithm becomes incorrect. This is exactly why a pure brute-force approach becomes expensive: $O(n^2)$. Because it blindly checks every possible pair.

With that said, the Divide and Conquer solution becomes clever by avoiding unnecessary comparisons.

Now let us go back to the actual plane. After recursively dividing the points:



- The closest pair on the left side is: $P[6, 8]$
- The closest pair on the right side is: $P[15, 16]$

Their corresponding distances are: δ_L and δ_R . Suppose: $\delta_L < \delta_R$. Then:

$$\delta = \min(\delta_L, \delta_R) = \delta_L$$

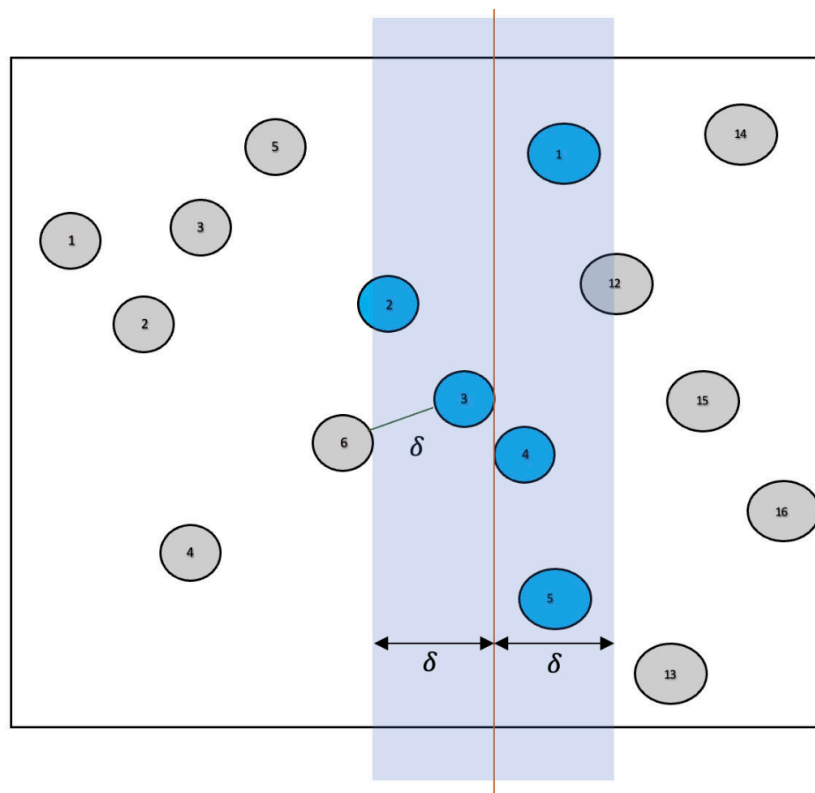
So δ becomes our current minimum distance. But again, we still have not checked cross-border pairs. And this is where the geometric insight appears.

Think carefully about this. Suppose two points from different halves are actually closer than δ . Can they be very far from the dividing line? No. Because if either point is farther than δ away from the middle line, then the distance between them must already exceed δ .

So we arrive at an important observation:

If a cross-border pair has distance smaller than δ , then both points must lie within distance δ from the middle dividing line.

This creates a vertical strip around the middle. Its total width becomes: 2δ

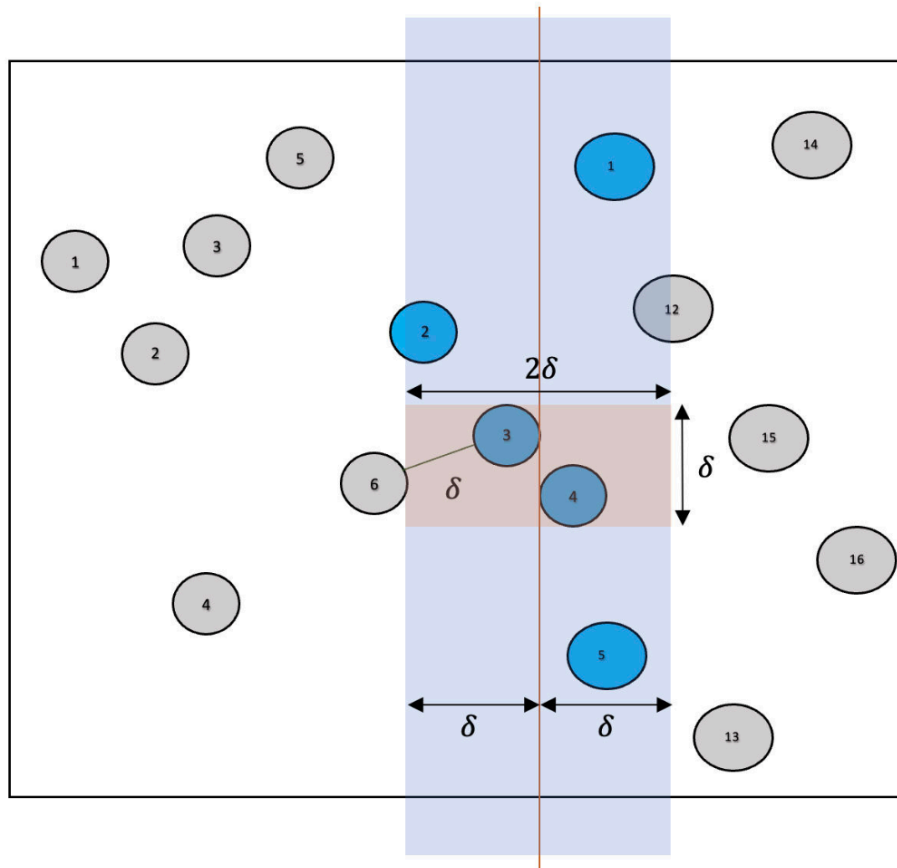


Only points inside this strip are considered further. And this changes everything. Instead of checking every pair in the plane, we only inspect a very small subset of points.

Now another clever optimization appears. Look closely at the image above: **The points inside the strip are sorted according to their y-coordinates instead of their x-coordinates.** Why? Because vertically nearby points are the only ones capable of producing smaller distances.

This dramatically reduces the number of comparisons. For example: $P[3, 1], P[3, 5], P[2, 1]$. These pairs are obviously too far apart vertically. So we immediately ignore them.

Here comes the clever part. Instead of comparing every point inside the strip with every other point, the algorithm scans through the strip one point at a time. For each point, we imagine a temporary rectangle of dimensions: $2\delta \times \delta$, and only examine nearby points that fall inside this rectangle.



As the scan moves through the strip, each candidate pair is compared against the current minimum distance δ . However, most comparisons are automatically discarded because if either point in the pair lies outside the current scanning rectangle of dimensions: $2\delta \times \delta$, then it is already guaranteed to be too far away—either horizontally or vertically—to produce a smaller distance than δ .

As a result, the algorithm only performs comparisons on nearby candidate points inside the rectangle. Inside one of these scanning rectangles, the only

meaningful pair becomes: $P[3, 4]$. Since their distance is smaller than the current δ , they become the new closest pair inside the strip.

And when we convert these strip indices back into the original plane points, we obtain: $P[8, 9]$, which becomes the overall closest pair in the plane.



Note: If the strip-scanning process does not discover any cross-border pair whose distance is smaller than δ , then that current δ (obtained from either δ_L or δ_R) is confirmed as the overall closest pair distance in the entire plane.

At first, checking all points inside the strip may still seem expensive. It almost looks like another brute-force step. But surprisingly, it is not. A geometric proof shows something remarkable:

For every point inside the strip, we only need to compare at most 7 points after it.

Because the strip is sorted by the y-coordinate, points farther away vertically cannot possibly produce a smaller distance. So the strip-processing step actually runs in: $O(n)$ time. This is the key optimization that makes the algorithm efficient.

Time Complexity Analysis - The Divide and Conquer algorithm recursively divides the problem into two halves.

The recurrence relation becomes:

$$T(n) = 2T\left(\frac{n}{2}\right) + O(n) + O(n \log n) + O(n)$$

Simplifying:

$$T(n) = 2T\left(\frac{n}{2}\right) + O(n \log n)$$

This gives an overall complexity of: $O(n(\log n)^2)$

However, there is an even better optimization.

If we presort all points according to the (y)-coordinate before recursion begins, then we no longer need to repeatedly sort the strip during recursive calls.

- This improves the final complexity to: $O(n \log n)$. Which is dramatically faster than the brute-force approach: $O(n^2)$
- The space complexity becomes: $O(\log n)$ due to recursive function calls.

Summary

The Closest Pair of Points problem is one of the most elegant applications of the Divide and Conquer strategy.

Instead of checking every possible pair blindly, we divide the plane into smaller halves, solve each half recursively, carefully examine only the important points near the dividing line

The most beautiful part of this algorithm is not just the recursion itself. It is the geometric observation that allows us to ignore most comparisons completely. And this transforms an expensive quadratic-time problem into an efficient $O(n \log n)$ algorithm.

References

<https://www.geeksforgeeks.org/dsa/closest-pair-of-points-using-divide-and-conquer-algorithm/>

https://www.youtube.com/watch?v=6u_hWxbOc7E